Prompt

I am designing an isolated virtual laboratory environment to be used for hands-on training and studying (cybersecurity and network orchestration). I need a precise CPU, RAM, and Storage allocation breakdown for 5 Linux Virtual Machines running on a single host hypervisor.

Lab Usage Details:
- Target Audience / Concurrency: Single student following the instructions from the lab guideline
- Usage Pattern: Short bursts of intensive scans during 1h of lab session, idle otherwise

VM Specifications:
- VM1 (Network Simulation & Viz): Mininet simulating 24 hosts and 7 OpenFlow switches + Plotly Dash web UI. (Traffic profile: low idle traffic).
- VM2 (IDS / Security Monitoring): Suricata IDS processing mirrored traffic from VM1. Minimal ruleset based on ET Open standard set and a few custom rules.
- VM3 (Attacker / Pentest Box): PHP/Apache/SQLite web app + Kali/offensive CLI tools (Nmap, Hydra, Hping3). Very low usage with on demand scans, brute-force and ping usage
- VM4 (SDN Orchestrator): Python web app running via Gunicorn + local MongoDB database. (Dataset size: minimal study data).
- VM5 (Vulnerable Target): Legacy Apache server + OpenSSL vulnerable to Heartbleed. (Usage: target for nmap vulnerability scan).

Please provide:
- Baseline Allocation Table: Minimal and Recommended vCPU, RAM (GB), and Disk space (GB) per VM for smooth operation.

Additional information

- Models and versions: ChatGP 5.6 Sol Light, Claude Sonnet 5, Gemini 3.6 Flash
- Disagreements were resolved by choosing the lowest resource requirement

ChatGPT

For a **single-student isolated laboratory**, the goal should be to allocate enough resources to avoid bottlenecks during short bursts of activity while keeping the overall host requirements modest. Since the workloads are mostly sequential (the student typically performs one activity at a time), CPU oversubscription is acceptable (approximately 1.5–2:1), while RAM should generally be allocated close to the VM's working set to avoid swapping.

Baseline Resource Allocation

VM	Purpose	Minimal vCPU	Recommended vCPU	Minimal RAM	Recommended RAM	Minimal Disk	Recommended Disk	Notes
VM1	Mininet + 24 hosts + 7 OpenFlow switches + Dash UI	2	4	4 GB	8 GB	20 GB	40 GB	Mininet mainly consumes CPU during topology startup and traffic generation. 24 hosts remain lightweight.

VM2	Suricata IDS	2	2	2 GB	4 GB	20 GB	40 GB	Small mirrored traffic and limited ruleset keep CPU usage relatively low. Extra RAM improves rule caching.
VM3	Kali + Apache/PHP/SQLite + Offensive tools	2	4	4 GB	8 GB	30 GB	60 GB	Kali packages occupy considerable disk space. Nmap and Hydra benefit from additional CPU during scans.
VM4	Python Web Application + Gunicorn + MongoDB	2	2	2 GB	4 GB	20 GB	40 GB	MongoDB dataset is minimal. CPU demand is

								generally low except during initialization.
VM5	Vulnerable Apache/OpenSSL Server	1	2	1 GB	2 GB	10 GB	20 GB	Very lightweight target machine.

Total Laboratory Requirements

Allocation	vCPU	RAM	Storage
Minimal	**9 vCPU**	**13 GB**	**100 GB**
Recommended	**14 vCPU**	**26 GB**	**200 GB**

Suggested Hypervisor Host

Considering virtualization overhead and leaving sufficient resources for the host operating system:

Host Size	CPU	RAM	Storage	Suitable For
Minimum Host	12 physical cores (or 6 cores / 12 threads)	24 GB	250 GB SSD/NVMe	Functional laboratory with occasional CPU contention during scans
Recommended Host	16 physical cores (or 8 cores / 16 threads)	32 GB	500 GB NVMe SSD	Smooth execution with minimal contention
Comfortable / Future	20–24 cores	64 GB	1 TB NVMe	Supports larger Mininet topologies, additional VMs, or

Growth				multiple concurrent experiments

Expected Resource Utilization

CPU

- **VM1** experiences the highest CPU usage while Mininet starts and during traffic generation.
- **VM3** briefly becomes CPU-intensive when running Nmap, Hydra, or Hping3.
- **VM2** continuously processes mirrored packets but, given the low traffic volume, rarely exceeds 20–30% utilization of two vCPUs.
- **VM4** and **VM5** remain mostly idle.

Consequently, it is uncommon for all VMs to reach peak CPU usage simultaneously.

Memory

Approximate working memory during normal lab execution:

VM	Typical Working RAM
VM1	3–5 GB
VM2	1.5–3 GB
VM3	3–5 GB
VM4	2–3 GB
VM5	<1 GB

Typical total memory consumption is therefore around **12–17 GB**, making a **32 GB host** sufficient while leaving room for the hypervisor and filesystem cache.

Storage

Storage requirements are driven primarily by the operating systems rather than the applications:

- Kali Linux: **25–35 GB**
- Ubuntu Server instances: **10–20 GB**
- Suricata logs: typically **<1 GB** per lab session
- MongoDB: only a few hundred MB
- Dash application: negligible
- PHP application: negligible

Allocating **200 GB** across the VMs leaves sufficient free space for snapshots, package updates, and temporary files.

Recommended Allocation for Educational Laboratories

For the best balance between performance and resource efficiency, the following configuration is recommended:

VM	vCPU	RAM	Disk
VM1 – Mininet	**4**	**8 GB**	**40 GB**
VM2 – Suricata	**2**	**4 GB**	**40 GB**
VM3 – Kali/Attacker	**4**	**8 GB**	**60 GB**
VM4 – SDN Orchestrator	**2**	**4 GB**	**40 GB**
VM5 – Vulnerable Target	**2**	**2 GB**	**20 GB**

Total: 14 vCPUs, 26 GB RAM, and 200 GB disk

This configuration provides enough headroom for realistic cybersecurity exercises—including Mininet emulation, IDS monitoring, web application hosting, vulnerability scanning, and SDN orchestration—without overprovisioning resources for a single-student laboratory. It also scales well if you later increase the Mininet topology (for example, to 50–100 hosts) or add more IDS rules and attack scenarios.

Gemini

Here is the precise resource allocation breakdown tailored specifically to your lab setup. Given your usage pattern—a single student conducting short, bursty sessions rather than running continuous heavy workloads—the hypervisor can rely on light resource overcommit (especially for vCPUs) while reserving dedicated RAM to prevent memory thrashing.

Baseline Allocation Table

VM ID & Role	Workload Highlights	Minimal vCPU	Rec. vCPU	Minimal RAM	Rec. RAM	Minimal Disk	Rec. Disk
VM1: Network Simulation & Viz	Mininet (24 hosts + 7 OF switches) + Plotly Dash UI	2	4	2 GB	4 GB	20 GB	30 GB
VM2: IDS / Security Monitoring	Suricata processing mirrored VM1 traffic	2	4	2 GB	4 GB	20 GB	40 GB
VM3: Attacker / Pentest Box	PHP/Apache/SQLite app + Kali / CLI offensive tools	1	2	2 GB	4 GB	25 GB	40 GB

VM4: SDN Orchestrator	Python (Gunicorn) + MongoDB	1	2	2 GB	4 GB	15 GB	25 GB
VM5: Vulnerable Target	Legacy Apache + OpenSSL (Heartbleed target)	1	1	1 GB	2 GB	10 GB	15 GB
TOTALS		**7**	**13**	**9 GB**	**18 GB**	**90 GB**	**150 GB**

Resource Sizing Justification

- **VM1 (Mininet & Plotly Dash):** Mininet lightweight host processes run inside lightweight Linux namespaces rather than full OS instances, keeping CPU overhead low during idle state. However, during burst traffic generation across 7 switches or while rendering real-time Plotly Dash graphs, multi-core scheduling and 4 GB RAM ensure smooth graph redraws and packet processing without dropped frames.
- **VM2 (Suricata IDS):** Suricata is multithreaded and benefits significantly from multiple vCPUs during active Nmap scans or Hydra brute-force bursts from VM3. 4 GB RAM provides enough space for Suricata's detection engine pattern matching tables and log buffers without disk swapping.
- **VM3 (Attacker Box):** Running offensive CLI tools like hping3 or multi-threaded hydra requires modest CPU bursts. 25–40 GB storage allows for a lean CLI/headless Kali installation with essential wordlists (e.g., rockyou.txt) and scan outputs.
- **VM4 (SDN Orchestrator):** Gunicorn workers and MongoDB are lightweight for small-scale lab datasets. MongoDB will comfortably stay within 1–2 GB RAM active working set, but 4 GB total headroom prevents out-of-memory (OOM) kills during Python app execution.
- **VM5 (Vulnerable Target):** Legacy Linux running basic Apache + vulnerable OpenSSL requires minimal resources. 1 vCPU and 1–2 GB RAM is plenty to sustain targeted Nmap vulnerability scans and exploit payloads without crashing the guest OS.

Hypervisor Requirements Summary

- **Host CPU:** A 4-core / 8-thread physical CPU is sufficient for the **Minimal** setup. An 8-core / 16-thread host will handle the **Recommended** setup seamlessly.
- **Host RAM:** Provision **16 GB** minimum on the hypervisor host (allowing ~9 GB for VMs + hypervisor overhead) or **24 GB / 32 GB** for the Recommended tier.
- **Storage Note:** Disk sizes assume thin provisioning. Thin provisioning allows you to overcommit storage capacity, as the VMs will only consume physical disk space as logs and datasets grow.

Claude

Here's a sizing breakdown tailored to a single-student, single-host lab with idle/burst usage — this lets you lean on modest baselines with some CPU oversubscription, since all 5 VMs won't peak simultaneously.

Per-VM Allocation Table

VM	Role	vCPU (Min / Rec)	RAM GB (Min / Rec)	Disk GB (Min / Rec)	Rationale
VM1	Mininet (24 hosts/7 switches) + Dash UI	2 / 4	2 / 4	20 / 40	Mininet uses lightweight network namespaces, not full VMs, so CPU/RAM cost per simulated node is low. Dash's web server + browser rendering (if headless plotting/caching) is the main RAM consumer, not the topology itself.
VM2	Suricata IDS (ET Open + custom rules)	2 / 4	4 / 8	20 / 40	Suricata's rule-matching engine (Hyperscan/AC) loads the full ET Open ruleset into memory at startup — this is RAM-hungry regardless of traffic volume. Multi-threaded detection benefits from extra cores during the 1h burst window. Extra disk headroom for eve.json/alert logs and PCAP retention.
VM3	Attacker/Pentest box (Kali tools + PHP/Apache/SQLite)	1 / 2	2 / 4	20 / 40	Nmap/Hydra/Hping3 are CPU-bursty but brief and single-threaded per scan; idle most of the session. Extra disk budgeted for wordlists (rockyou, SecLists), Nmap NSE scripts, and scan output logs.

VM4	SDN Orchestrator (Gunicorn + MongoDB)	2 / 2	2 / 4	20 / 30	MongoDB's WiredTiger storage engine reserves ~50% of available RAM for cache by default, even with a tiny dataset — so RAM floor is driven by Mongo, not your app logic. Gunicorn workers are lightweight for a single-user lab (1-2 workers is enough).
VM5	Vulnerable target (legacy Apache + OpenSSL/Heartbleed)	1 / 1	1 / 2	15 / 20	Deliberately minimal legacy stack; it only needs to sit idle and respond to scan/exploit traffic. No real workload beyond serving HTTPS handshakes.
Host Total	—	**8 / 13**	**11 / 22**	**95 / 170**	Sum of columns above.

Notes on sizing logic

CPU oversubscription is safe here. Since only one student drives one VM at a time (attacker scans VM5, or watches Mininet/Dash while Suricata listens passively), you can allocate a host with **6–8 physical cores** and still hit the "Recommended" column via oversubscription — true simultaneous 100% load across all 5 VMs won't happen in this workflow.

RAM is the tighter constraint, mainly because of Suricata's ruleset loading and MongoDB's cache reservation — both allocate memory upfront regardless of actual traffic/data volume. Don't undersize these two below the "Minimal" column or you'll see OOM kills or Suricata falling back to slower matching.

Disk recommendations assume ephemeral lab use (rebuild/reset between sessions). If you want persistent PCAP archives or long-term Suricata log retention across many sessions, bump VM2 to 60–80GB.

Suggested host baseline: 8 vCPU / 24GB RAM / 200GB disk (SSD strongly preferred — Suricata and MongoDB are both I/O-sensitive) comfortably covers the "Recommended" column with headroom for the hypervisor itself (ESXi/Proxmox/KVM overhead).

Want me to also break down a **network/vSwitch topology diagram** (port groups, VLANs for isolating VM3↔VM5 attack traffic from the mirrored feed to VM2) to go with this?